

Task sheet Week 8: How to read a research paper? k232001

Task: Review three empirical research papers (you already have from assignment 1) related to your domain or area of interest.

Structured notes for each paper using the template below.

Author	Year	Title	Keywords	Research Context	Research Problem	Contribution	Method	Key findings
Wei Hung Pan, Ming Jie Chok, Jonathan Leong Shan Wong, Yung Xin Shin, Yeong Shian Poon, Zhou Yang, Chun Yong Chong, David Lo, and Mei Kuan Lim	2024	Assessing AI Detectors in Identifying AI-Generated Code: Implications for Education	Software Engineering Education, AI-Generated Code, AI-Generated Code Detection	Students increasingly use Large Language Models (LLMs) to generate programming code, prompting educators to rely on Artificial Intelligence Generated Content (AIGC) detectors to prevent academic dishonesty	While AIGC detectors are proven effective on natural language text, their ability to reliably detect the complex syntax of programming code remains uncertain, risking unfair grading	The first comprehensive empirical study evaluating 5 AIGC detectors against 13 specific variants of AI-generated code prompts, compiling a dataset of 5,069 samples per variant	Collected human-written Python solutions and used ChatGPT to generate AI equivalents under 13 prompt variations (e.g., removing stopwords, adding assertions); evaluated detectors using metrics like accuracy, True Positive Rate, and True Negative Rate	Existing detectors perform poorly (around 50% accuracy) at distinguishing human from AI-generated code. Specific variants, such as asking the AI to "mimic a human response" or adding assertion test codes, severely degrade detection capabilities

Maria Trigka and Elias Dritsas	2025	The Evolution of Generative AI: Trends and Applications	Generative AI, multi-modal learning, ethical AI and governance, multi-modal AI	Generative AI has rapidly evolved through architectures like VAEs, GANs, Diffusion Models, and Transformers, enabling unprecedented synthetic content creation across text, images, and structured data	The rapid integration of Generative AI poses significant technical, ethical, and societal challenges (such as bias, deepfakes, computational inefficiency, and lack of interpretability) that lack unified governance and structural solutions	Provides a structured, comprehensive taxonomy and structural roadmap connecting foundational architectures to domain-specific applications, emerging trends, and ethical mitigation strategies	A taxonomy-driven literature review surveying peer-reviewed journals and conferences to categorize generative models, evaluate their performance trade-offs, and map them to real-world deployment challenges	Future generative systems require scalable architectures (like Mixture-of-Experts), multi-modal integration, and Reinforcement Learning, but must be strictly coupled with robust AI governance, explainability frameworks, and fairness-aware training to mitigate societal risks like misinformation and labor displacement
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Qinghua Lu, Liming Zhu, Xiwei Xu, Zhenchang Xing, and Jon Whittle	2024	Toward Responsible AI in the Era of Generative AI: A Reference Architecture for Designing Foundation Model-Based Systems	Foundation Models, System Architecture, Responsible AI, Reference Architecture (inferred from content and title)	Foundation Models (FMs) are becoming the core building blocks of modern AI systems, gradually absorbing the responsibilities of traditional external software components.	There is a lack of systematic, system-level architectural guidance on how to manage the "moving boundary" of FM integration while ensuring accountability, trustworthiness, and risk mitigation in AI systems	Proposes an evolutionary model for AI architecture and provides a pattern-oriented, "responsible-by-design" reference architecture for FM-based systems	Qualitative architectural analysis that identifies seven core design decisions (e.g., using a chain of FMs vs. a single ultra-large FM; passive vs. proactive interaction) and empirically validates the reference architecture by mapping it to a real-world RAI chatbot system	Architecture is shifting from "FM-as-a-connector" to monolithic FM setups. To maintain adaptability and responsibility, systems should use a three-layer architecture (System, Operation, Supply Chain) implementing patterns like continuous risk assessors, verifiers, microkernels, and black box recorders
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